The reasoning behind this conclusion is so subtle that it took physicists more than three decades to appreciate fully. But after covering the essential features of quantum mechanics we will see that the core of the argument reduces to nothing more complex than a Click and Clack puzzler.

Casting a Wave

If you shine a laser pointer on a little piece of black, overexposed 35mm film from which you have scratched away the emulsion in two extremely close and narrow lines, you will see direct evidence that light is a wave. If you've never done this, it's worth a try (you can use many things in place of the film, such as the wire mesh in a fancy coffee plunger). The image you will see when the laser light passes through the slits on the film and hits a screen consists of light and dark bands, as in Figure 4.1, and the explanation for this pattern relies on a basic feature of waves. Water waves are easiest to visualize, so let's first explain the essential point with waves on a large, placid lake, and then apply our understanding to light.

A water wave disturbs the flat surface of a lake by creating regions where the water level is higher than usual and regions where it is lower than usual. The highest part of a wave is called its *peak* and the lowest part is called its *trough*. A typical wave involves a periodic succession: peak followed by trough followed by peak, and so forth. If two waves head toward each other—if, for example, you and I each drop a pebble into the lake at nearby locations, producing outward-moving waves that run into each other—when they cross there results an important effect known as *interference*, illustrated in Figure 4.2a. When a peak of one wave and a peak of the other cross, the height of the water is even greater, being the sum of the two peak heights. Similarly, when a trough of one wave and a trough of the other cross, the depression in the water is even deeper, being the sum of the two depressions. And here is the most important combination: when a peak of one wave crosses the trough of another, they tend to cancel each other out, as the peak tries to make the water go up while the trough tries to drag it down. If the height of one wave's peak equals the depth of the other's trough, there will be